Literature Review

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Paper Title

A Comparative Analysis of Traditional and Deep Learning Time Series Architectures for Influenza A Infectious Disease Forecasting

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Domain

Healthcare / Time-Series Forecasting

Research Problem

Influenza A causes an estimated 290,000–650,000 respiratory deaths annually worldwide, while official CDC surveillance reports are typically delayed by at least one week. The paper investigates whether modern deep learning sequence models can forecast Influenza

A case counts more accurately than traditional statistical time-series models to support timely public health planning and resource allocation.

Dataset

The study uses monthly Influenza A case counts covering strains such as A(H1N1)2009, A(H3N2), and A(H7N9) from January 2009 to December 2023. The data were obtained from *Our World in Data*. The first 168 months (2009–2022) were used for training, while the final 12 months (2023) were reserved for testing.

Methodology

The authors first analyzed the time series using statistical techniques including:

- Seasonal decomposition
- Mann-Kendall trend test
- Kruskal-Wallis seasonality test
- KPSS stationarity test

Two traditional forecasting models were developed:

- Holt-Winters Exponential Smoothing (ETS)
- Seasonal ARIMA

The residuals were validated using:

- Ljung-Box Test
- Lilliefors Test
- Goldfeld-Quandt Test

Six deep learning models were then trained:

- Simple RNN

- LSTM
- GRU
- BiLSTM
- BiGRU
- Transformer

Hyperparameters such as hidden units, activation functions, optimizers, learning rate, batch size, and dropout were optimized using Grid Search.

Model Architecture

- Simple RNN, LSTM, and GRU consist of four stacked recurrent layers with 64 hidden units and 30% dropout.
- BiLSTM and BiGRU use bidirectional recurrent layers with the Adam optimizer.
- Transformer is implemented in PyTorch with:
  - Embedding dimension = 64
  - 4 Encoder Layers
  - 3 Attention Heads
  - Feed Forward Dimension = 128
  - 30% Dropout

Results

The study found that all deep learning models outperformed traditional statistical approaches.

The Transformer achieved the best overall performance, while GRU outperformed both Simple RNN and LSTM despite having a simpler architecture.

Model	Test MSE
Simple RNN	0.0602
LSTM	0.0545
GRU	0.0486
Transformer	0.0433

Table 1: Performance Comparison of Deep Learning Models

Limitations

- Monthly data are too coarse for weekly public health decisions.
- The 12-month testing period is relatively small.
- Only historical case counts are used; external factors such as weather, mobility, and vaccination rates are ignored.
- Deep learning models require greater computational resources than traditional methods.
- Prediction intervals and uncertainty estimation are not provided.

Suggested Future Improvements

- Include environmental and demographic variables.
- Develop probabilistic forecasting with prediction intervals.
- Use weekly surveillance data instead of monthly observations.
- Explore hybrid models such as SARIMA-LSTM.
- Validate models in real-time disease surveillance systems.

Critical Analysis

This paper presents a comprehensive comparison between statistical forecasting techniques and modern deep learning architectures. A major strength is its rigorous methodology, beginning with statistical analysis before applying deep learning models. The comparison across eight forecasting approaches provides valuable insight into the strengths and weaknesses of each model.

Although the Transformer achieved the highest accuracy, the GRU demonstrated competitive performance while maintaining a simpler architecture and lower computational cost. The study therefore suggests that GRU offers an effective balance between efficiency and predictive performance.

One limitation is the relatively short testing period and the absence of uncertainty estimation, which reduces its applicability for real-world healthcare decision-making. Future research should incorporate additional external variables and real-time forecasting systems.

Reference

Agyemang, E. F., Rodrigo, H., & Agbenyeavu, V. (2025).
A Comparative Analysis of Traditional and Deep Learning Time Series Architectures for Influenza A Infectious Disease Forecasting.
arXiv:2507.19515.